

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech. (CBCS) Degree Examinations

December 2022

MA1002-1 – CALCULUS AND DIFFERENTIAL EQUATIONS

(Common to AI&ML/CCE/CSE/ISE/AI&DS)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer type Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The angle ϕ between the radius vector and tangent for the vector for the curve $r = ae^{\theta \cot \alpha}$ is
 A) $\tan \alpha$ B) $\cot \alpha$
 C) α D) θ
- The radius of curvature of the curve $y = e^x$ at the point where it crosses the y-axis is
 A) $2\sqrt{2}$ B) $\sqrt{2}$
 C) 2 D) $\sqrt{2}/2$
- The function $f(x) = x^2$ satisfy the Rolle's theorem in which of the following intervals
 A) $[1, 2]$ B) $[0, 1]$
 C) $[-1, 1]$ D) none of these
- The derivative of $\arcsin \frac{dy}{dx}$ for the curve $x = f(y)$ is
 A) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ B) $1 + \left(\frac{dy}{dx}\right)^2$
 C) $1 + \left(\frac{dx}{dy}\right)^2$ D) $\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$
- If $f(x, y, z) = x^2 + xyz + z$, then the value of f_x at $(1, 1, 1)$ is
 A) 0 B) 1
 C) 3 D) -1
- If $z = 3x^2y + 5y$ then $\frac{\partial^2 z}{\partial x \partial y}$ is
 A) $3xy$ B) $6x$
 C) $3x + 5$ D) $6xy$
- Which of the following is correct?
 A) $\frac{\partial}{\partial x} \left(\tan^{-1} \frac{y}{x} \right) = \frac{-y}{x^2 + y^2}$ B) $\frac{\partial}{\partial x} \left(\tan^{-1} \frac{y}{x} \right) = \frac{x}{x^2 + y^2}$
 C) $\frac{\partial}{\partial x} \left(\tan^{-1} \frac{y}{x} \right) = \frac{x-y}{x^2 + y^2}$ D) $\frac{\partial}{\partial x} \left(\tan^{-1} \frac{y}{x} \right) = \frac{x+y}{x^2 + y^2}$
- The function $f(x, y) = x^2 + y^2 + 6x - 12$ has an extreme value at which one of the following intervals
 A) $(-3, 0)$ B) $(0, 3)$
 C) $(0, 0)$ D) $(-3, -3)$
- If $\phi = x^3 + y^3 - 2z$ then $\nabla \phi$ at $(1, -1, 1)$ is
 A) $3\hat{i} + 3\hat{j} - 2\hat{k}$ B) 4
 C) $3\hat{i} + 3\hat{j}$ D) $\hat{i} + \hat{j} - \hat{k}$
- $\vec{F}$ is said to be solenoidal if
 A) $\nabla \cdot \vec{F} = 0$ B) $\nabla \cdot \vec{F} = 0$
 C) $\nabla \times \vec{F} = \vec{0}$ D) $\nabla (\nabla \cdot \vec{F}) = 0$

11. If $f(x, y) = 2xy + y^2$ then Hessian of f is
 A) $\begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$
 C) $\begin{bmatrix} 0 & 0 \\ 2 & 2 \end{bmatrix}$ D) $\begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$
12. If $\vec{F} = ax\hat{i} + b\hat{j}$ where a, b are constants, is irrotational, then which of the following is true?
 A) $\nabla \times \vec{F} = a$ B) $\nabla \times \vec{F}$ is a scalar quantity
 C) $\nabla \times \vec{F} = 0$ D) $\nabla \times \vec{F} \neq 0$
13. If two roots of the auxiliary equation of a second order linear differential equation with constant coefficients are real and equal, then the complementary solution is of the form
 A) $y_c = Ae^{m_1x} + Be^{m_2x}$ B) $y_c = Ae^{mx} + Bxe^{mx}$
 C) $y_c = Ae^{mx} + Be^{mx}$ D) $y_c = Ae^m + Bxe^m$
14. Which of the following function is not a solution of the differential equation $y'' + y = 0$?
 A) $y = \sin 2x$ B) $y = 2 \sin x$
 C) $y = \cos x$ D) $y = 15 \cos x$
15. The partial differential equation of the expression $z = ax + by$, where a and b are arbitrary constants, is
 A) $z = px - qy$ B) $z = p + q$
 C) $z = pq$ D) $z = px + qy$
16. Which of the following equation is a parabolic partial differential equation?
 A) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ B) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
 C) $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ D) $\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial^2 u}{\partial t^2}$
17. The value of $\int_0^1 \int_0^2 xy \, dx \, dy =$
 A) 0 B) 1
 C) 3 D) -3
18. If R is the region bounded by the circles $r = 1$, $r = 2$ and the lines $\theta = 0$ and $\theta = \pi/2$ then $\iint_R r \, dr \, d\theta =$
 A) $\int_0^{\pi/2} \int_0^2 r \, dr \, d\theta$ B) $\int_0^{\pi/2} \int_0^1 r \, dr \, d\theta$
 C) $\int_0^{\pi/2} \int_1^2 r \, dr \, d\theta$ D) $\int_0^{\pi/2} \int_2^\infty r \, dr \, d\theta$
19. If $\int_0^1 \int_0^2 \int_0^3 ax \, dz \, dy \, dx = \int_0^1 \int_0^2 \int_0^3 by \, dz \, dy \, dx$ then
 A) $a = 2b$ B) $b = 2a$
 C) $a = b$ D) $a = -b$
20. Which of the following is true?
 A) $\Gamma(n+1) = n\Gamma(n)$ B) $\Gamma(n) = n\Gamma(n+1)$
 C) $\Gamma(n+1) = (n+1)\Gamma(n)$ D) $\Gamma(n) = (n+1)\Gamma(n+1)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) State and prove the Cauchy's Mean value theorem.	6	L1	1	1
	b) A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction.	6	L2	2	2
	c) If $x = r \cos \theta$ and $y = r \sin \theta$, find $J = \frac{\partial(x, y)}{\partial(r, \theta)}$	4	L1	2	1
2.	a) With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L1	1	2
	b) Obtain the expansion of e^{xy} in powers of $(x-1)$ and $(y-1)$ up to second degree terms.	6	L2	2	2

- c) Prove that the curves $r = a \cos \theta$ and $r = a \sin \theta$ intersect orthogonally. 4 L2 2 1
3. a) If ρ_1, ρ_2 be the radii of curvature at the extremities of any chord of the cardioid $r = a(1 + \cos \theta)$ which passes through the pole, show that $\rho_1^2 + \rho_2^2 = 16 \frac{a^2}{9}$. 6 L2 1 1
- b) If $u = f(y - z, z - x, x - y)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. 6 L2 2 2
- c) Obtain the Maclaurin's Series expansion of the function $\cos x$. Expand up to three non-vanishing terms. 4 L1 1 1

Unit – II

4. a) If $\phi = x^3 + y^3 + z^3 - 3xyz$, find $\vec{F} = \nabla \phi$. Hence find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$. Evaluate these at the point $(1, -1, 1)$. 6 L1 3 1
- b) Solve the differential equation: $(D^3 + D)y = 1 + x^3$ 6 L2 4 1
- c) Find the value of b such that
 $A = (bx^2y + yz)\hat{i} + (xy^2 - xz^2)\hat{j} + (2xyz - 2x^2y^2)\hat{k}$
 is solenoidal. 4 L1 3 2
5. a) Evaluate the gradient and Hessian of the function $f(x, y) = x \sin(y)$ at the point $(1, \pi/2)$. 6 L1 3 1
- b) Solve the differential equation: $(D^3 + 6D^2 + 9D)y = e^{-3x} + 2$ 6 L2 4 2
- c) Form partial differential equations by eliminating the arbitrary constants/functions:
 i) $z = ax^2 + by^2$ ii) $z = f(x^2 + y^2)$ 4 L2 4 1
6. a) Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$. 6 L2 3 2
- b) Solve the P.D.E. by the method of separation of variables:
 $u_x = 2u_t + u$ given that $u(x, 0) = 6e^{-3x}$. 6 L1 4 2
- c) Find the solution of the differential equation:
 $y'' - 3y' + 2y = 0$, given that $y(0) = 0$ & $y'(0) = 1$. 4 L1 4 1

Unit – III

7. a) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar co-ordinates. 6 L2 5 1
- b) Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 6 L2 5 2
- c) Evaluate in terms of beta function: $\int_0^{\pi/2} \sin^7 \theta \cos^5 \theta d\theta$ 4 L1 5 1
8. a) Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. 6 L1 5 1
- b) Evaluate $\int_0^\infty x^4 e^{-x^2} dx$ in terms of Gamma functions. 6 L2 5 2
- c) Evaluate the triple integral: $\int_0^1 \int_0^2 \int_1^2 x^2 yz dx dy dz$. 4 L1 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
